Lecture 11

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1 Rationality of Square Roots (Continued Proof)

1.1 Theorem: Rational Square Roots of Integers

Let $n \in \mathbb{N}$. Then $\sqrt{n}$ is rational if and only if n is a perfect square.

1.1.1 Proof

We use the Fundamental Theorem of Arithmetic (Unique Prime Factorization).

Let the prime factorization of n be:

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

Forward Direction ($\Leftarrow$): If n is a perfect square, then $n = a^2$ for some $a \in \mathbb{Z}$. Thus $\sqrt{n} = a \in \mathbb{Q}$.

Reverse Direction ($\Rightarrow$): Suppose $\sqrt{n} \in \mathbb{Q}$. Then we can write $\sqrt{n} = \frac{a}{b}$ where $a, b \in \mathbb{Z}, b \neq 0$, and $\gcd(a, b) = 1$ (fraction is in simplest form).

Squaring both sides:

$$n = \frac{a^2}{b^2} \implies n \cdot b^2 = a^2$$

Let the prime factorizations be:

$$n = \prod p_i^{\alpha_i}, \quad b = \prod q_j^{\beta_j}, \quad a = \prod r_l^{\gamma_l}$$

Substituting these into the equation $n \cdot b^2 = a^2$:

$$\left(\prod p_i^{\alpha_i}\right) \cdot \left(\prod q_j^{\beta_j}\right)^2 = \left(\prod r_l^{\gamma_l}\right)^2$$

$$\prod p_i^{\alpha_i} \cdot \prod q_j^{2\beta_j} = \prod r_l^{2\gamma_l}$$

By the Fundamental Theorem of Arithmetic, the exponent of any prime factor on the Left Hand Side (LHS) must equal the exponent of that prime on the Right Hand Side (RHS).

- The exponents on the RHS ($2\gamma_l$) are all **even**.
- The exponents coming from b^2 on the LHS ($2\beta_j$) are all **even**.
- Therefore, the exponents from n (α_i) must also be **even** to maintain equality.

Since all α_i are even, let $\alpha_i = 2k_i$. Then:

$$n = p_1^{2k_1} \dots p_k^{2k_k} = (p_1^{k_1} \dots p_k^{k_k})^2$$

Thus, n is a perfect square.

1.1.2 Example

Let $n = 3^2 \cdot 5^7$. Is $\sqrt{n}$ rational?

- The exponent of 5 is 7 (odd).
- Therefore, n is not a perfect square, and $\sqrt{n}$ is irrational.

2 Modular Arithmetic

2.1 Introduction: Parity Arithmetic

Before defining general modular arithmetic, consider the arithmetic of Even (e) and Odd (o) numbers.

$+$	e	o
e	e	o
o	o	e

$\times$	e	o
e	e	e
o	e	o

This arithmetic corresponds to calculations modulo 2.

2.2 Definition: Congruence Modulo m

Let $a, b, m \in \mathbb{Z}$ with $m > 1$. We say a is congruent to b modulo m if m divides the difference $a - b$.

$$a \equiv b \pmod{m} \iff m \mid (a - b)$$

2.2.1 Properties

Congruence is an equivalence relation. For $m > 1$:

1. **Reflexive**: $a \equiv a \pmod{m}$ (since $m \mid 0$).
2. **Symmetric**: $a \equiv b \pmod{m} \iff b \equiv a \pmod{m}$.
3. **Transitive**: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

2.3 Arithmetic Operations

If $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$, then:

1. $a + b \equiv c + d \pmod{m}$
2. $a \cdot b \equiv c \cdot d \pmod{m}$

2.3.1 Example: Clock Arithmetic

Consider time modulo 12 ($m = 12$).

- $9:00 + 5:00 = 14:00$.
- $14 \equiv 2 \pmod{12}$.
- So, $9 + 5 \equiv 2 \pmod{12}$.

2.4 Proofs of Arithmetic Properties

2.4.1 Addition Property

Claim: $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m} \implies a + b \equiv c + d \pmod{m}$.
Proof: By definition:

$$a - c = k_1 m$$

$$b - d = k_2 m$$

Adding the equations:

$$(a - c) + (b - d) = k_1 m + k_2 m$$

$$(a + b) - (c + d) = m(k_1 + k_2)$$

Since $k_1 + k_2$ is an integer, $m \mid ((a + b) - (c + d))$. Therefore, $a + b \equiv c + d \pmod{m}$.

2.4.2 Multiplication Property

Claim: $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m} \implies ab \equiv cd \pmod{m}$.
Proof: We write $a = c + mk_1$ and $b = d + mk_2$.

$$ab = (c + mk_1)(d + mk_2)$$

$$ab = cd + c(mk_2) + d(mk_1) + m^2 k_1 k_2$$

$$ab - cd = m(ck_2 + dk_1 + mk_1 k_2)$$

Since the RHS is a multiple of m, $m \mid (ab - cd)$. Therefore, $ab \equiv cd \pmod{m}$.

2.4.3 Power Property

By induction on the multiplication property: If $a \equiv b \pmod{m}$, then $a^n \equiv b^n \pmod{m}$ for $n \in \mathbb{N}$.

3 Applications and Problems

3.1 Problem 1: Calculating Large Powers

Calculate the remainder of 6^{30} when divided by 5, 7, and 13.

3.1.1 Case 1: Modulo 5

$$6 \equiv 1 \pmod{5}$$
$$6^{30} \equiv 1^{30} \equiv 1 \pmod{5}$$

3.1.2 Case 2: Modulo 7

$$6 \equiv -1 \pmod{7}$$
$$6^{30} \equiv (-1)^{30} \equiv 1 \pmod{7}$$

3.1.3 Case 3: Modulo 13

We compute powers of 6 modulo 13 until we find a pattern (usually 1 or -1):

- $6^1 \equiv 6$
- $6^2 = 36 = 2 \cdot 13 + 10 \equiv 10 \equiv -3 \pmod{13}$
- $6^3 \equiv 6 \cdot (-3) = -18 \equiv -5 \pmod{13}$ (Note: $-18 = -2 \cdot 13 + 8$, so also 8)
- $6^4 \equiv 6 \cdot (-5) = -30 \equiv -4 \equiv 9 \pmod{13}$
- $6^5 \equiv 6 \cdot (-4) = -24 \equiv 2 \pmod{13}$
- $6^6 \equiv 6 \cdot 2 = 12 \equiv -1 \pmod{13}$

Using $6^6 \equiv -1 \pmod{13}$:

$$6^{30} = (6^6)^5 \equiv (-1)^5 \equiv -1 \equiv 12 \pmod{13}$$

3.2 Problem 2: Diophantine Equation

Solve $x^2 - 3y = 2$ in integers.

3.2.1 Solution

We analyze the equation modulo 3.

$$x^2 - 3y = 2 \implies x^2 \equiv 2 \pmod 3$$

We check all possible values of $x \pmod 3$ to see if $x^2 \equiv 2$ is possible.

$x \pmod 3$	$x^2 \pmod 3$
0	0
1	1
2	$4 \equiv 1$

Observation: The square of an integer modulo 3 is always 0 or 1. It is never 2. **Conclusion**: There are no integer solutions for $x^2 - 3y = 2$.